Modeling the Dynamics of Alzheimer's Disease

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What Alzheimer's Disease Is

Alzheimer's Disease (AD) is a neurodegenerative disease characterized by many complex interactions that affect over 50 million people worldwide. Many different aspects of the disease are anticipated, but it is not yet known how these factors interact [2].

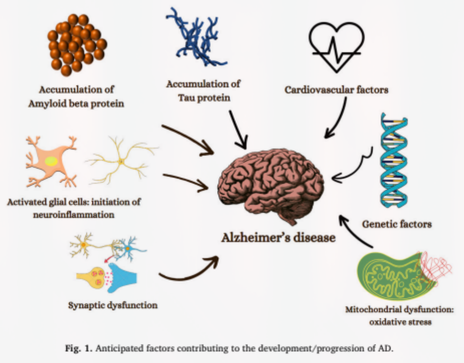

Figure: Various factors of AD from the literature

Hypotheses Galore

Various hypotheses have been proposed in order to understand the underlying mechanisms of AD; where the leading hypothesis for many years has been the *Amyloid Cascade Hypothesis*. However, there has been increasing support in the notion that it is insufficient in giving an effective description and treatment towards AD [1].

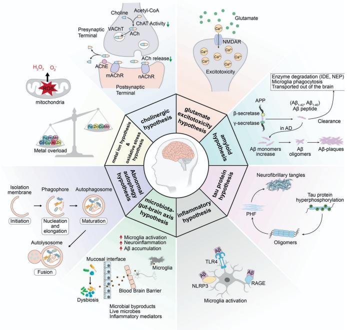

Figure: A visualization of the various hypotheses relating to the mechanisms of AD

Formation of a Newly Proposed Model

In order to understand these factors and subsequently the behavior of AD, many systems of ordinary differential equations (ODEs) have been proposed related to these hypotheses; but, a few are highlighted in regard to creating a newly proposed dynamical system:

- Observing the dynamics between A_β and Ca^{2+} [3].
- Using causal models observing factors of genetic, non-amyloid dependent, and biomarker cascades [4].

The Proposed Model

Due to the feedback loop created by A_β and Ca^{2+}, along with the interaction between the hyperphosphorylation of τ-protein caused by A_β, a proposed causal model is described by the following system of ODEs:

$$\frac{dA_\beta}{dt} = a_1 + \underbrace{a_2 \frac{Ca}{Ca+\sigma}}_{\text{Holling Type II}} - r_1 A_\beta \tag{1}$$

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \tag{2}$$

$$\frac{d\tau}{dt} = c_1 + \underbrace{c_2 A_\beta + c_3 Ca}_{A_\beta,\ \text{Ca}^{2+}\ \text{interaction}} - r_3 \tau \tag{3}$$

$$\frac{dN}{dt} = d_1 + d_2 \tau - k_4 N \tag{4}$$

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C, \tag{5}$$

where $r_n = k_n + u_n$, and u_n is the treatment parameter.

On Invariance and Boundedness

The system can be shown to be *positively invariant*. This is shown by

$$\left.\frac{dA_\beta}{dt}\right]_{A_\beta=0}, \left.\frac{dCa}{dt}\right]_{Ca=0}, \left.\frac{d\tau}{dt}\right]_{\tau=0}, \left.\frac{dN}{dt}\right]_{N=0}, \left.\frac{dC}{dt}\right]_{C=0}.$$

Since the model follows biological scenarios, non-negative parameter choices mean that each compartment is non-negative, and thus the non-negative orthant is invariant.

The system can also be shown to be *bounded* by

$$a_1 + a_2\frac{Ca}{Ca+\sigma} - r_1A_\beta \leq a_1 + a_2 - r_1A_\beta < 0 \text{ if } A_\beta > \frac{a_1+a_2}{r_1}$$
$$b_1 + b_2A_\beta - r_2Ca \leq b_1 + b_2M - r_2Ca < 0 \text{ if } Ca > \frac{b_1+b_2M}{r_2}.$$

since A_β, Ca feed into the other compartments it follows that each subsequent compartment is also bounded

Discussion of Proposed Model

When analyzing the model, while equilibria are intractable, global asymptotic stability has been determined for the model. To show this, consider the first two compartments that are uncoupled from the rest of the proposed model:

$$\frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca}{Ca+\sigma} - r_1 A_\beta$$
$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca$$

For showing the number of equilibria, solve $dA_\beta/dt = 0$ in terms of A_β. Substituting this into dCa/dt and simplifying (using CAS), a quadratic is obtained of the form

$$-(r_1 r_2)Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma)Ca + a_1 b_2 \sigma + b_1 r_1 \sigma = 0.$$

Using *Descartes' Rule of Signs*, one positive root is determined, and hence a unique positive real equilibrium (A_β^*, Ca^*).

Discussion of Proposed Model

Then to show the non-existence of limit cycles, consider $\mathbf{F}(A_\beta, Ca) = \left\langle \frac{dA_\beta}{dt}, \frac{dCa}{dt} \right\rangle$, $\alpha(A_\beta, Ca) = 1$. It can be shown that

$$\nabla \cdot \mathbf{F} = -r_1 - r_2 < 0 \implies \nabla \cdot (\alpha \mathbf{F}) = -r_1 - r_2 < 0.$$

Since $\nabla \cdot (\alpha \mathbf{F}) < 0$, by the Bendixson-Dulac Theorem there exists no limit cycles. This shows that the unique equilibrium for the uncoupled system is globally asymptotically stable.

Now, using (A_β^*, Ca^*), notice that

$$0 = c_1 + c_2 A_\beta + c_3 Ca - r_3 \tau$$
$$\tau^* = \frac{c_1 + c_2 A_\beta^* + c_3 Ca^*}{r_3}.$$

since a globally stable equilibrium has been determined for the uncoupled system, it follows that τ will also tend toward some equilibrium that is globally stable. This provides a similar cascade to N, C, and thus there exists a unique globally asymptotically stable equilibrium for the proposed model.

Visualization of Model

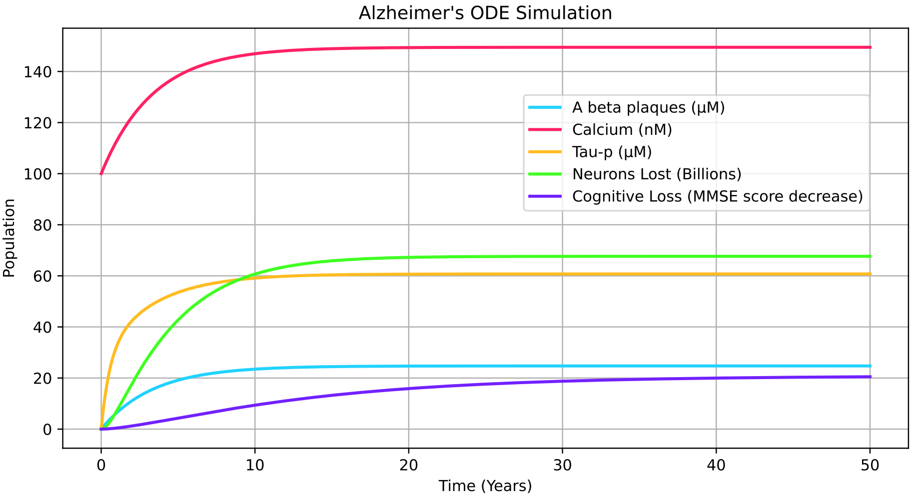

Figure: Visuzaliation of compartments in proposed ODE model.

Sensitivity Analysis

Using parameters obtained from the literature, Sobol Sensitivity analysis is performed using ± 0.15 of the respective parameter value in order to obtain a parameter range. The table below provides a quantitative representation of the most sensitive parameters in the model (values closer to zero signifies lesser sensitivity).

Params	k5	k3	e3	k1	c1	c2	...	c3	...
Sobol 1st Order	0.2761	0.2743	0.1191	0.0579	0.0527	0.0463	...	0.0047	...
Sobol Total Order	0.2776	0.2760	0.1225	0.0596	0.0549	0.0478	...	0.0047	...

Table: Quantitative Representation of Sensitivity Analysis

Sensitivity Analysis

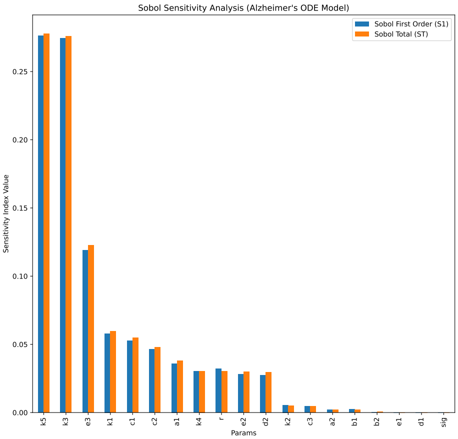

Figure: Visualization of Sobol Orders

Sensitivity Analysis

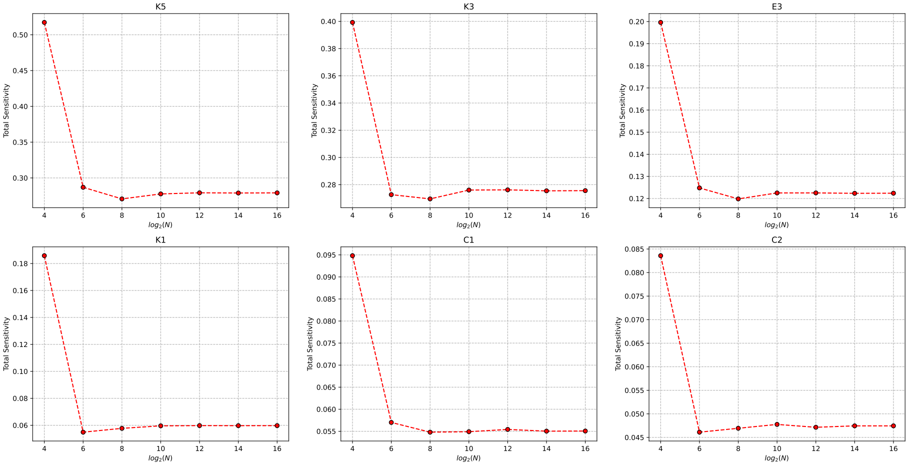

Figure: 6 most sensitive parameters across $N \in [2^4, 2^{16}]$

Optimal Control

Recall the system of equations:

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2\frac{Ca}{Ca+\sigma} - k_1A_\beta - u_1A_\beta \\ \frac{dCa}{dt} = b_1 + b_2A_\beta - k_2Ca - u_2Ca \\ \frac{d\tau}{dt} = c_1 + c_2A_\beta + c_3Ca - k_3\tau - u_3\tau \\ \frac{dN}{dt} = d_1 + d_2\tau - k_4N \\ \frac{dC}{dt} = e_1 + e_2NR + e_3\tau - k_5C \end{cases}$$

The purpose of optimal control is to determine the best way to administer treatments with minimal side effects (cost). The optimization problem is to minimize the objective function,

$$J(u_1, u_2, u_3) = \int_0^K A_1N + A_2C + A_3u_1^2 + A_4u_2^2 + A_5u_3^2\,dt,$$

where u_n are non-negative treatment rates for each corresponding drug. In optimal control, each u_n will be a function of time $u_n(t)$, so treatment levels can vary.

Future Directions

In order to provide a more interesting description of the mechanisms of AD, future directions include:

- Begin working with ADNI data in order to provide application modeled to real-world scenarios.
- Apply optimal control to simulations to find values for treatments.

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Closing

Thank you. Questions?